

Assignment - 2.2

Control Flow Testing - A white box testing approach

1 Write a program to find a given number is prime number or not.

→ read n, i
if $n \leq 1$

print("not prime")
else

$i = 2$

while $i < n$

if $n \% i == 0$

print("not prime")

else

$i = i + 1$

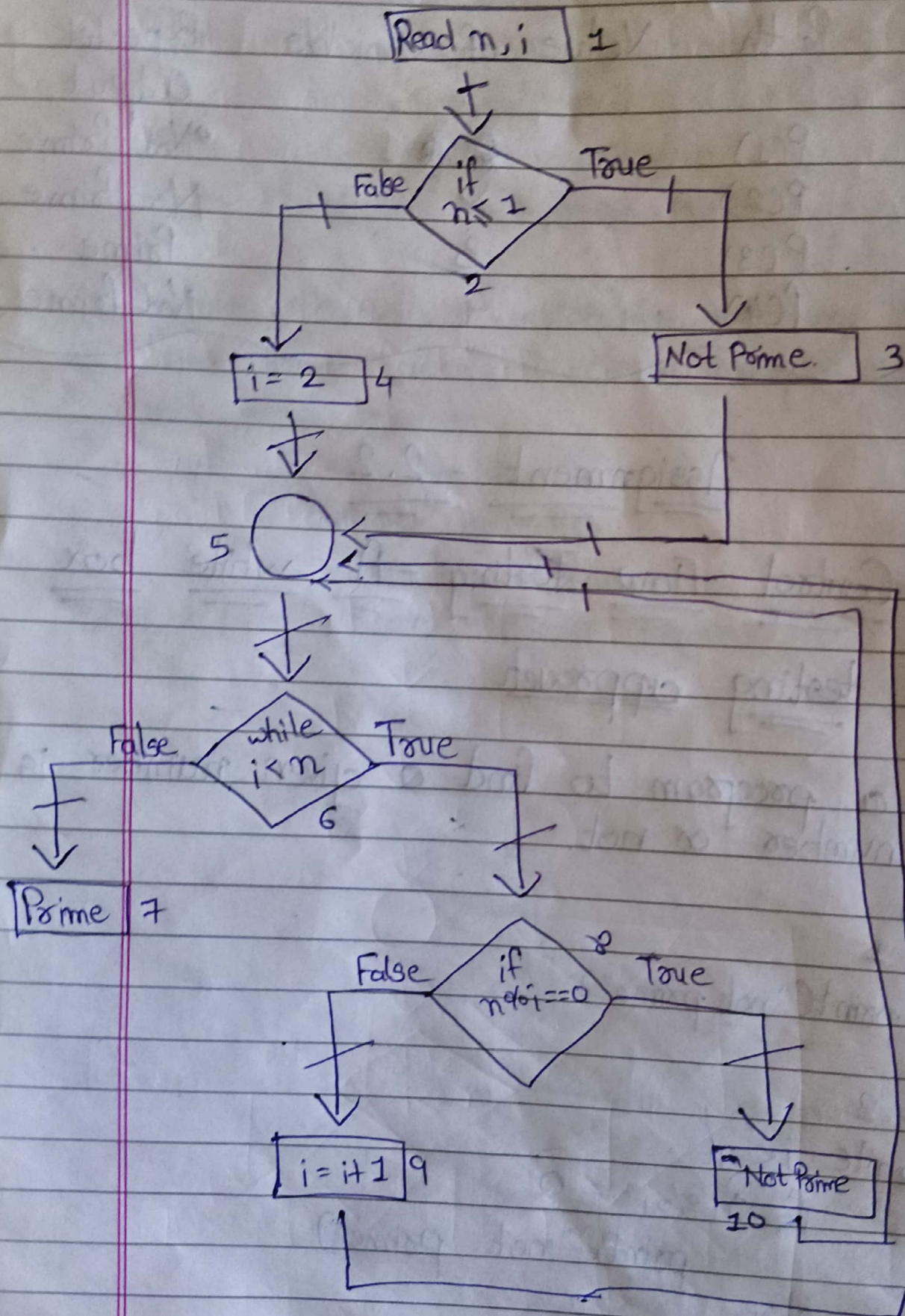
end if

end while

print("Prime")

end if

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→ Nodes = 10
Edges = 12

$$\begin{aligned} \rightarrow C(G) &= E - N + 2 \\ &= 12 - 10 + 2 \\ &= 2 + 2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} \rightarrow C(G) &= P + 1 \\ &= 3 + 1 \\ &= 4 \end{aligned}$$

→ Path:-

1] 1-2T-3-5-6F-7

2] 1-2F-4-5-6T-8T-10-5-6F-7

3] 1-2F-4-5-6T-8F-9-5-6F-7

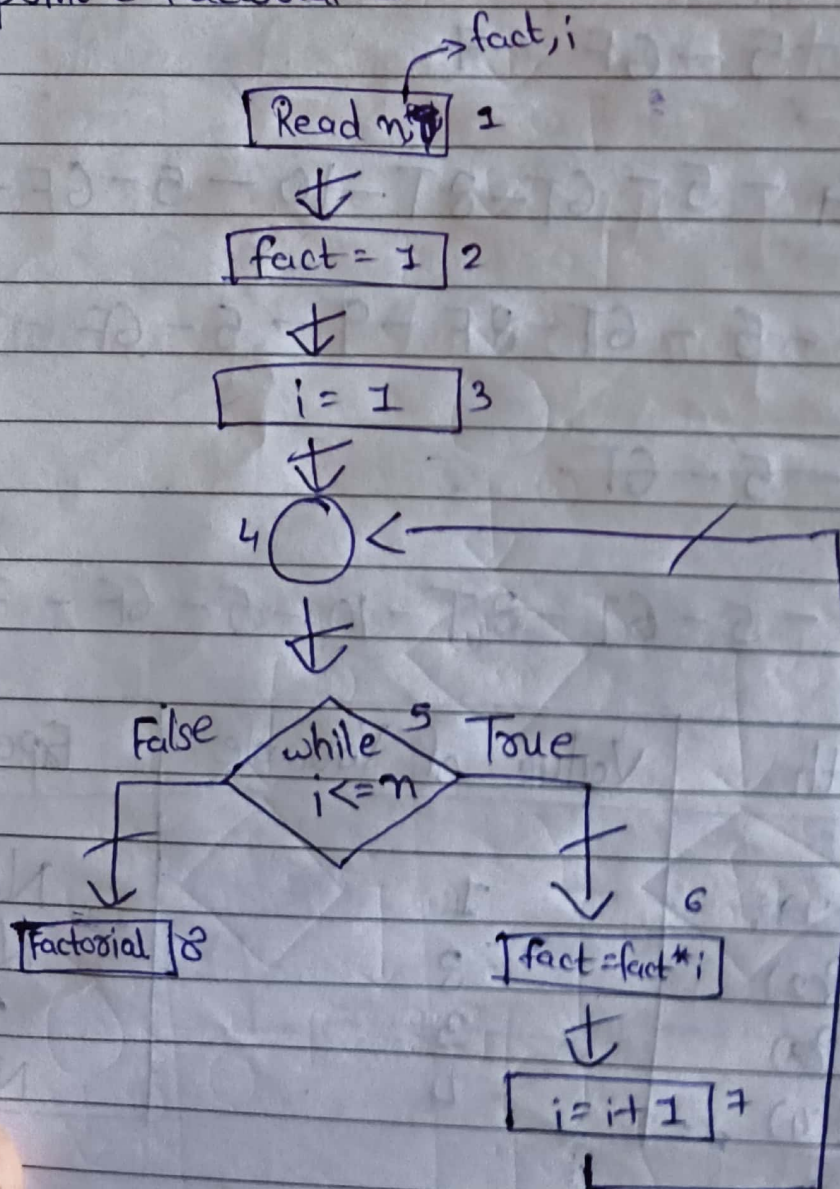
4] ~~1-2^T-4-5-6T~~

4] 1-2T-3-5-6T-8T-10-5-6F-7

→ Ss.no.	Path	Value of Variable	Expected Output
1	Pc1)	1	Not Prime
2	Pc2)	8	Not Prime
3	Pc3)	3	Prime
4	Pc4)	4	Not Prime

2 Write a program to find the factorial of a given number.

→ Read n
fact = 1
i = 1
while i ≤ n
 ~~fact~~ fact = fact * i
 i = i + 1
end while
print ("Factorial")



→ Nodes = 8
Edges = 8

$$\begin{aligned} \rightarrow (CG) &= E - N + 2 \\ &= 8 - 8 + 2 \\ &= 0 + 2 \\ &= 2 \end{aligned}$$

$$\begin{aligned} \rightarrow C(G) &= P + 1 \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

→ Path :-

1] 1 - 2 - 3 - 4 - 5T - 6 - 7 - 4 - 5F - 8

2] 1 - 2 - 3 - 4 - 5F - 8

Sr.No.	Path	Value of Variable	Expected Output
1	Pc1)	3	6
2	Pc2)	4	24

3 Write a program to search a given number from an array. If number is found, then give the index of the number ~~is~~ if number is not found return -1.

→ Read n, array a[n], key, i, found

i = 0

found = 0

while i < n

if a[i] = key

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print C"Element found at index",i)
found = 1

end if

i = i + 1

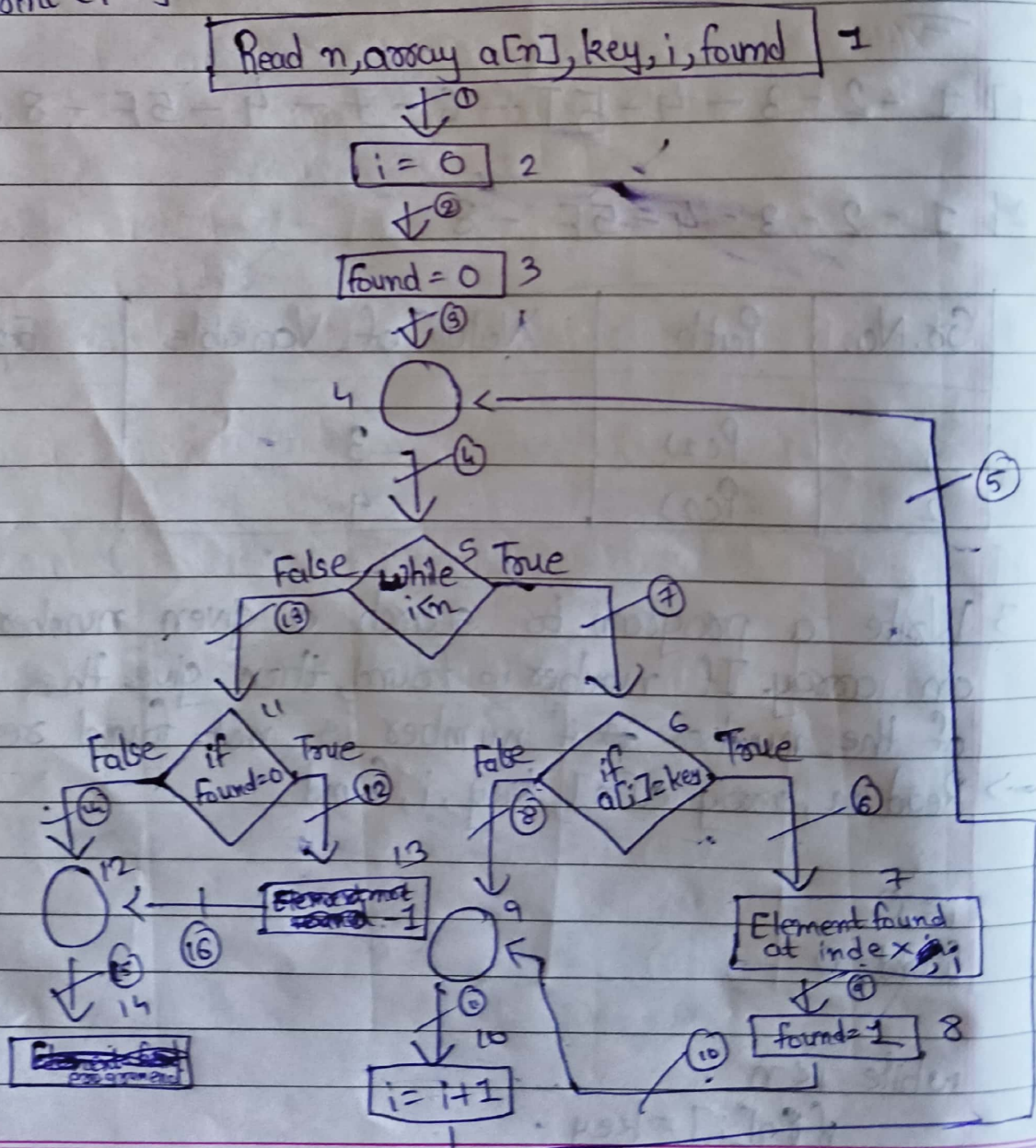
end while

if found = 0

print C"Element not found")

end if

print (program end)



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→ Node = 14

→ Edges = ~~35~~ 16

$$\begin{aligned} \rightarrow (C_G) &= E - N + 2 \\ &= 16 - 14 + 2 \\ &= 2 + 2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} \rightarrow (C_G) &= P + 1 \\ &= 3 + 1 \\ &= 4 \end{aligned}$$

Path

1] 1-2-3-4-5T-6T-7-8-9-10-4-5F-11F-12-14

2] 1-2-3-4-5T-6F-9-10-4-5F-11T-13-12-14

3] 1-2-3-4-5F-11T-13-12-14

4] 1-2-3-4-5F-11F-12-14

array ~~a[5]~~ = a = [2, 3, 4, 6, 8, 9, 10, 12]

Sr. no.	Path	Value of Variable	Expected Output
1	Pc1)	2	Element found at index program end
2	Pc2)	11	-1 Element not found program end.
3	Pc3)	0	-1 element not found program end
4	Pc4)	-1	program end.

4 Write a program to find the average values from a given array falling in a given range specified by ~~Min~~^{MIN} and ~~Max~~^{MAX}.

→ Read n , array $a[n]$, MIN , MAX , i , sum , $count$

$i = 0$

$sum = 0$

$count = 0$

while $i < n$

if $a[i] \geq MIN$ and $a[i] \leq MAX$

$sum = sum + a[i]$

$count = count + 1$

end if

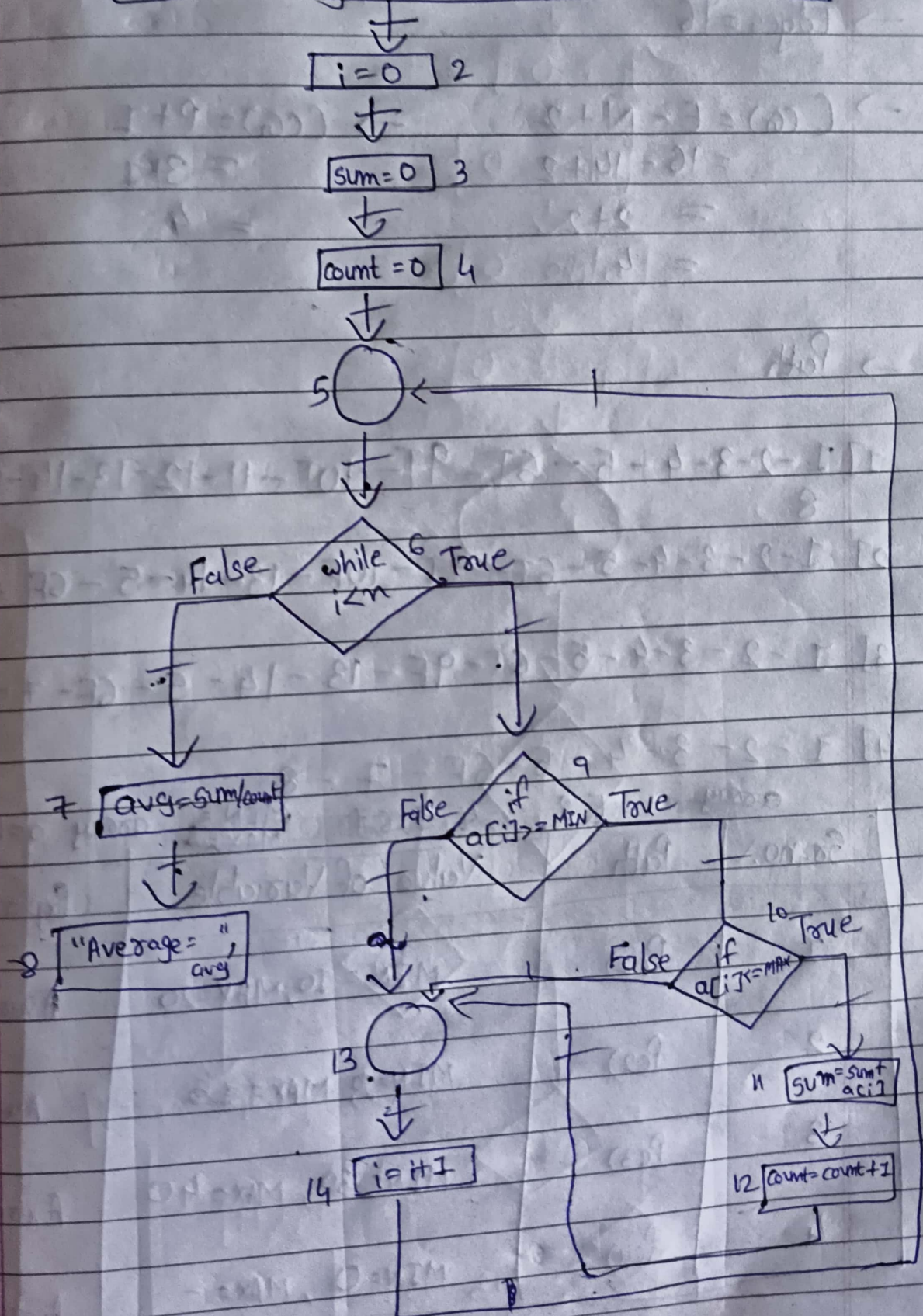
$i = i + 1$

end while

$avg = sum / count$

print ("Average = ", avg)

Read n , array $a[n]$, MIN, MAX, i , sum, count



→ Nodes = 14

→ Edges = 16

$$\begin{aligned}
 \rightarrow C(G) &= E - N + 2 \\
 &= 16 - 14 + 2 \\
 &= 2 + 2 \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 \rightarrow C(G) &= P + 1 \\
 &= 3 + 1 \\
 &= 4
 \end{aligned}$$

→ Path

1] 1-2-3-4-5-6T-9T-10T-11-12-13-14-5-6F-7-8

2] 1-2-3-4-5-6T-9T-10F-13-14-5-6F-7-8

3] 1-2-3-4-5-6T-9F-13-14-5-6F-7-8

4] 1-2-3-4-5-6F-7-8

~~array a = [2, 3, 4, 5, 6, 7]~~ MIN = MAX

~~array a = [5, 10, 15, 20, 25]~~

Sr. no.	Path	Value of Variable	Expected Output
1	Pc1)	MIN = 10, MAX = 20	Average = 15
2	Pc2)	MIN = 10, MAX = 15	Average = 12.5
3	Pc3)	MIN = 30, MAX = 40	Average = 0
4	Pc4)	MIN = 0, MAX = -	Average = 0

5 Write a program to find total number of digits, alphabets and special characters from a given string.
 → Read string s, i, digit, alpha, special

i = 0

digit = 0

alpha = 0

special = 0

while s[i] != NULL

if s[i] is alphabet

alpha = alpha + 1

else if s[i] is digit

digit = digit + 1

else

special = special + 1

end if

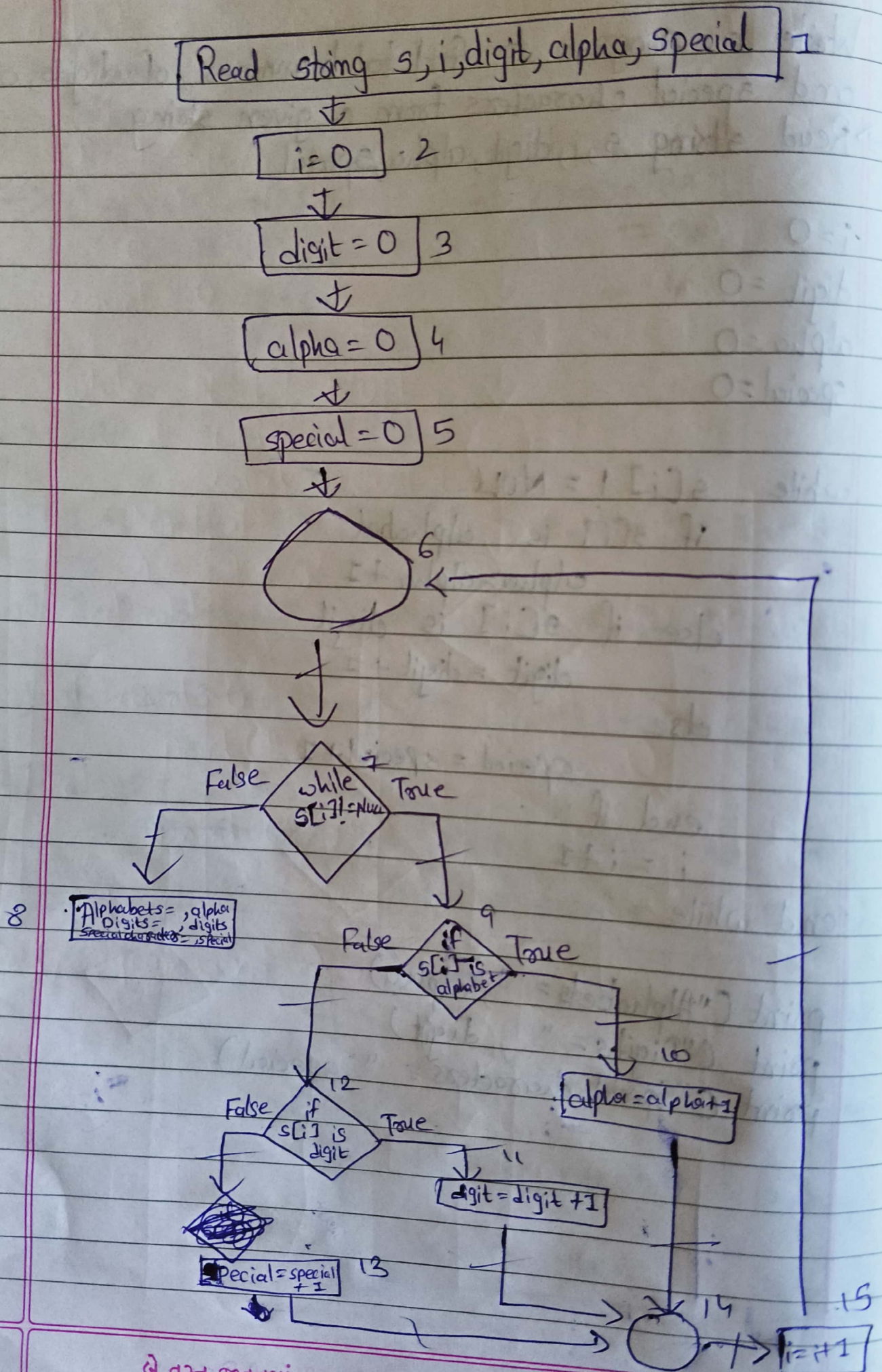
i = i + 1

end while

print ("Alphabets = ", alpha)

print ("Digits = ", digit)

print ("Special characters = ", special)



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→ Nodes = 15

→ Edges = 17

$$\begin{aligned} \rightarrow C(G) &= E - N + 2 \\ &= 17 - 15 + 2 \\ &= 2 + 2 \\ &= 4 \end{aligned}$$

$$\begin{aligned} \rightarrow C(G) &= P + 1 \\ &= 3 + 1 \\ &= 4 \end{aligned}$$

→ Path:-

1] 1-2-3-4-5-6-7T-9T-10-14-15-6-7F-8

2] 1-2-3-4-5-6-7T-9F-12T-11-14-15-6-7F-8

3] 1-2-3-4-5-6-7T-9F-12F-13-14-15-6-7F-8

4] 1-2-3-4-5-6-7F-8

So. no.	Path	Value of Variable	Expected Output
1	Pc1)	"ABC" alpha=3, digit=0, special=0	Alphabets=3 Digits=0 Special=0 characters
2	Pc2)	"123"	Alphabets=0 Digits=3 Special=0 characters
3	Pc3)	"@#&"	Alphabets=0 Digits=0 Special=3 characters
4	Pc4)	" " (empty string)	Alphabets=0 Digits=0 Special=0 characters

6. Write a program to accept a string and character
count occurrence of character if found else return
-1

→ Read String s, character ch, i, count

i = 0

count = 0

while s[i] != NULL

if s[i] == ch

count = count + 1

end if

i = i + 1

end while

if count > 0

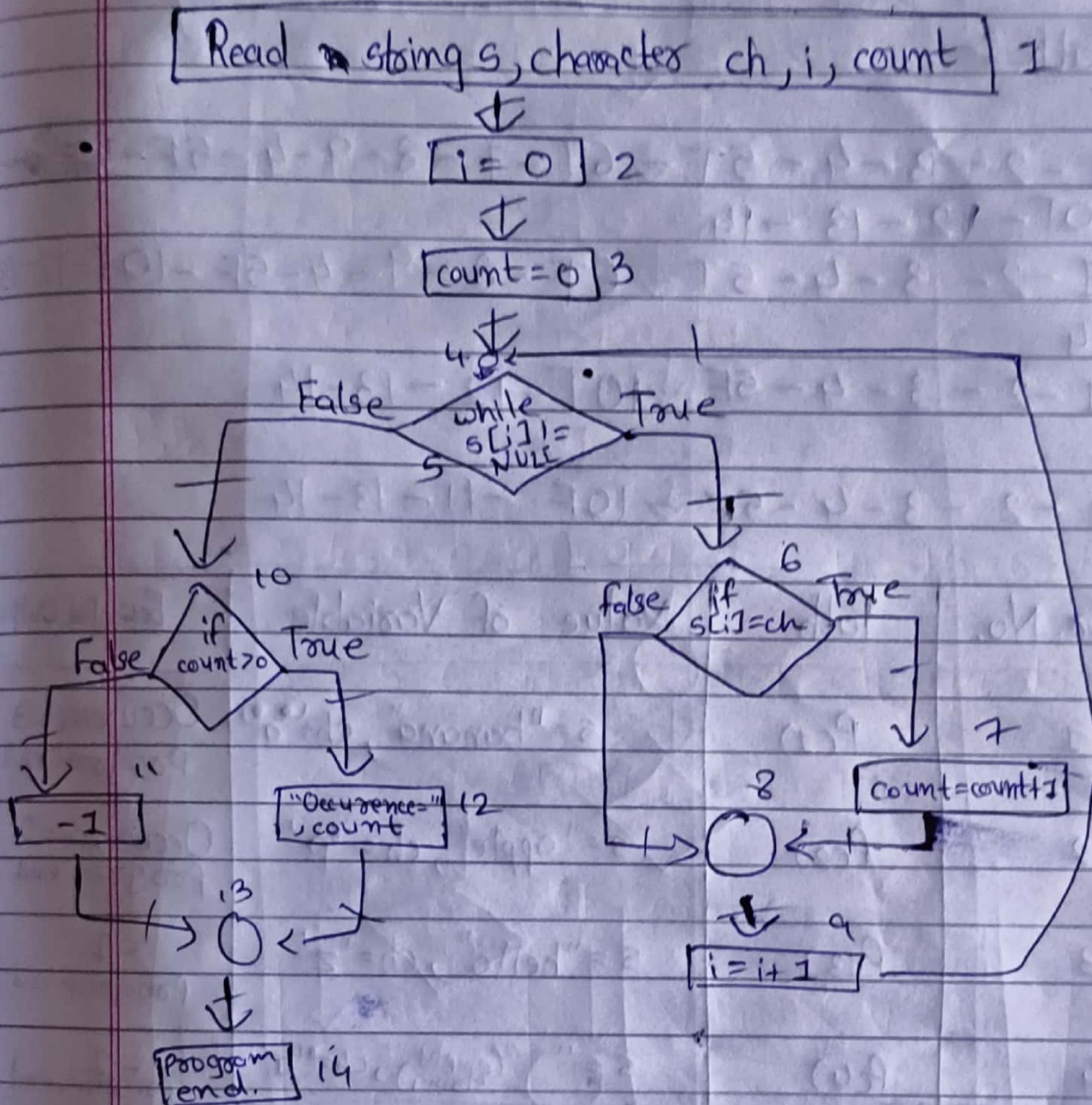
print("Occurrence = ", count)

else

print("-1")

end if

print("program end")



→ Nodes = 14

→ Edges = 16

→ $CCG = E - N + 2$

$$= 16 - 14 + 2$$

$$= 2 + 2$$

$$= 4$$

→ $CCG = P + 1$

$$= 3 + 1$$

$$= 4$$

→ Path :-

- 1] 1-2-3-4-5T-6T-7-8-9-4-5F-10T-11-12-13-14
- 2] 1-2-3-4-5T-6F-7-8-9-4-5F-10T-11-12-13-14
- 3] 1-2-3-4-5F-10T-11-12-13-14
- 4] 1-2-3-4-5F-10F-11-12-13-14

Sr. No.	Path	Value of Variable	Expected Output
1	Pc1)	s="banana", ch='a'	Occurrence=3 Program end
2	Pc2)	s="apple", ch='p'	Occurrence=2 Program end
3	Pc3)	s="hello", ch='z'	-1 Program end
4	Pc4)	s=" (empty string), ch='a'	-1 Program end

7 Write a program to return count of elements which are divisible by 4, 6, 7, 8 from an array of 10 elements.

→ Read array a[10], i, c4, c6, c7, c8

i = 0

c4 = 0

c6 = 0

c7 = 0

c8 = 0


```
while i < 10
```

```
    if a[i] % 4 == 0
```

```
        c4 = c4 + 1
```

```
    end if
```

```
    if a[i] % 6 == 0
```

```
        c6 = c6 + 1
```

```
    end if
```

```
    if a[i] % 7 == 0
```

```
        c7 = c7 + 1
```

```
    end if
```

```
    if a[i] % 8 == 0
```

```
        c8 = c8 + 1
```

```
    end if
```

```
    i = i + 1
```

```
end while
```

```
print("Divisible by 4 = ", c4)
```

```
print("Divisible by 6 = ", c6)
```

```
print("Divisible by 7 = ", c7)
```

```
print("Divisible by 8 = ", c8)
```


Read array $a[20]$, $i, c4, c6, c7, c8$ 1

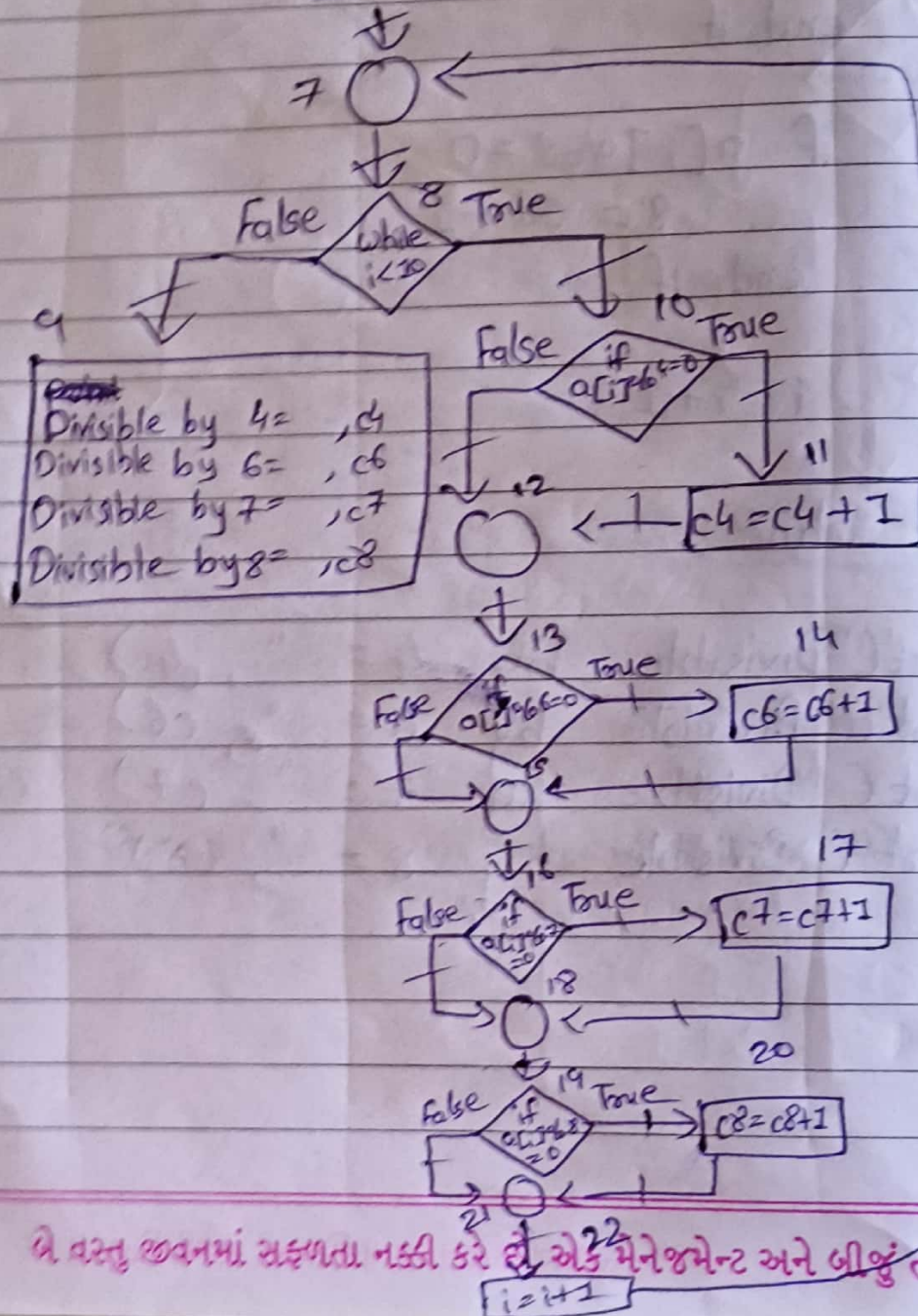
$i = 0$ 2

$c4 = 0$ 3

$c6 = 0$ 4

$c7 = 0$ 5

$c8 = 0$ 6



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→ Nodes = 22

→ Edges = 26

→ $CCG = E - N + 2$

$$= 26 - 22 + 2$$

$$= 4 + 2$$

$$= 6$$

→ $CCG = P + 1$

$$= 5 + 1$$

$$= 6$$

→ Path :-

1] 1-2-3-4-5-6-7-8T-10T-11-12-13T-14-15-16T-17-18-19T-20-21-22-7-8F-9

2] 1-2-3-4-5-6-7-8T-10T-11-12-13T-14-15-16T-17-18-19F-20-21-22-7-8F-9

3] 1-2-3-4-5-6-7-8T-10T-11-12-13T-14-15-16F-17-18-19F-20-21-22-7-8F-9

4] 1-2-3-4-5-6-7-8T-10T-11-12-13F-14-15-16T-17-18-19F-20-21-22-7-8F-9

5] 1-2-3-4-5-6-7-8T-10F-11-12-13F-14-15-16T-17-18-19T-20-21-22-7-8F-9

6] 1-2-3-4-5-6-7-8F-9

So. No.	Path	Value of Variable	Expected Output
1	Pc1)	$a = \{4, 8, 12, 16, 20, 24, 28, 32, 36, 40\}$	Divisible by 4 = 10 Divisible by 6 = 2 Divisible by 7 = 1 Divisible by 8 = 5
2	Pc2)	$a = \{6, 12, 18, 24, 30, 36, 42, 48, 54, 60\}$	Divisible by 4 = 3 Divisible by 6 = 10 Divisible by 7 = 1 Divisible by 8 = 1
3	Pc3)	$a = \{7, 14, 21, 28, 35, 42, 49, 56, 63, 70\}$	Divisible by 4 = 2 Divisible by 6 = 1 Divisible by 7 = 10 Divisible by 8 = 1
4	Pc4)	$a = \{8, 16, 24, 32, 40, 48, 56, 64, 72, 80\}$	Divisible by 4 = 10 Divisible by 6 = 2 Divisible by 7 = 1 Divisible by 8 = 10
5	Pc5)	$a = \{1, 2, 3, 5, 9, 10, 11, 13, 15, 17\}$	Divisible by 4 = 0 Divisible by 6 = 0 Divisible by 7 = 0 Divisible by 8 = 0

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6	P(6)	$a = \{4, 6, 8, 12, 14, 16, 21, 24, 28\}$	Divisible by $4 \neq 6$, Divisible by $6 = 3$, Divisible by $7 = 3$, Divisible by $8 = 2$
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8 Perform CFD for following snippets

→

Read ax, by

$ax = by - 10$

If $by \% 7 == 0$ then
print("ax")

else

print("by")

end if

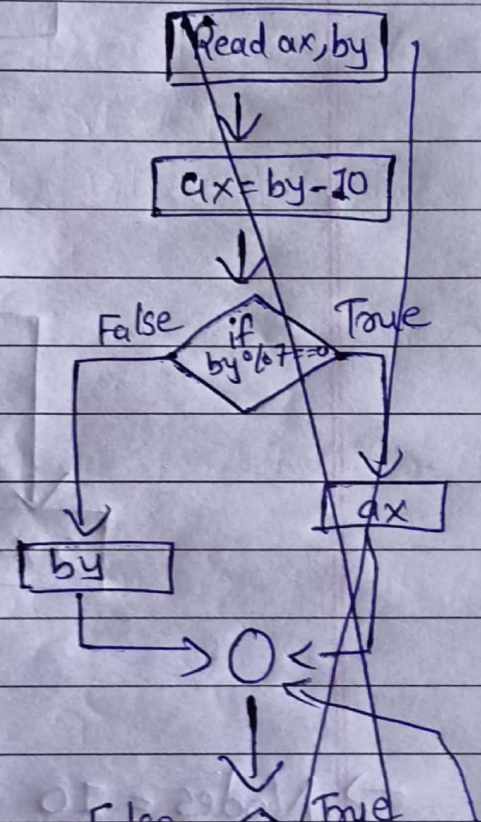
while $(ax \leq 5)$

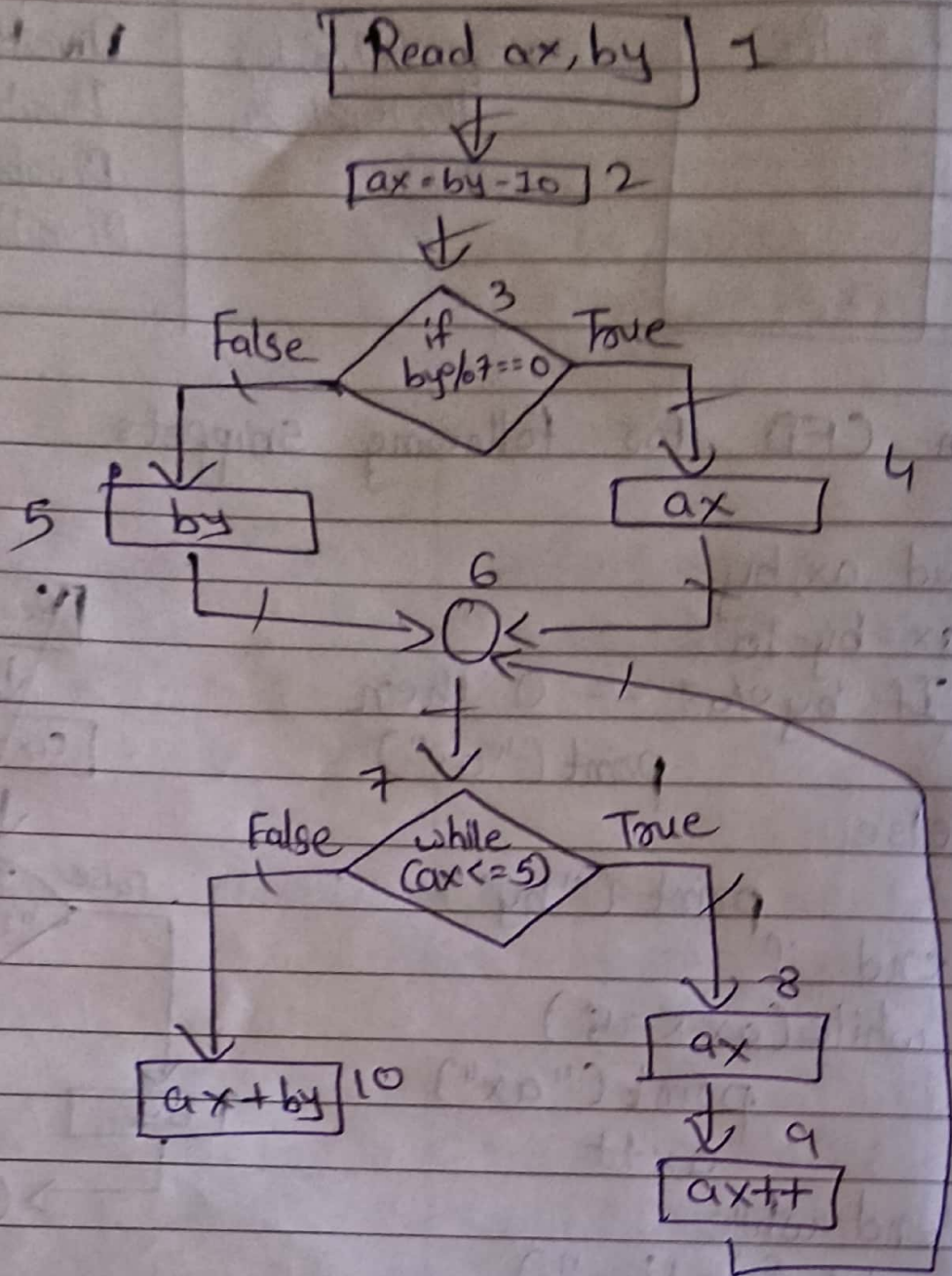
print("ax")

$ax++$

end while

print("ax+by")





→ Nodes = 10

→ Edges = 11

$$\begin{aligned}
 \rightarrow C(G) &= E - N + 2 \\
 &= 11 - 10 + 2 \\
 &= 1 + 2 \\
 &= 3
 \end{aligned}$$

$$\begin{aligned}
 \rightarrow C(G) &= P + 1 \\
 &= 2 + 1 \\
 &= 3
 \end{aligned}$$

→ Path :-

1] 1 - 2 - 3T - 4 - 6 - 7T - 8 - 9 - 6 - 7F - 10

2] 1 - 2 - 3T - 4 - 6 - 7F - 10

3] 1 - 2 - 3F - 5 - 6 - 7T - 8 - 9 - 6 - 7F - 10

Srs. No.	Path	Value of Variable	Expected Output
1	Pc1)	by = 14, ax = 4	4, 4, 5, 20
2	Pc2)	by = 21, ax = 11	11, 32
3	Pc3)	by = 12, ax = 2	12, 2, 3, 4, 5, 18